KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

CASE STUDY REPORT ON

LEAN MANUFACTURING PRACTICES IN SMALL AND CRAFT INDUSTRIES

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LIST OF ABBREVIATIONS

KU	Kathmandu University
JIT	Just In Time
TPM	Total Productive Maintenance
TPS	Toyota Production System
TTC	Technical Training Centre

CHAPTER 1 RATIONALE

1.1 Introduction to Lean Manufacturing

Lean manufacturing is a production approach originally developed in the United States to improve efficiency, reduce waste, and enhance responsiveness to customers and suppliers. It is a systematic production philosophy aimed at maximizing customer value while minimizing waste. The core idea of lean is to do more with less i.e., less time, less inventory, less space, less labor, and less cost without compromising quality. Any activity that does not add value from the customer's perspective is considered waste and should be reduced or eliminated. Lean focuses on identifying and eliminating non-value-adding activities from the production system.

Lean manufacturing emphasizes efficiency, flexibility, and customer responsiveness. Unlike traditional mass production which relies on large batch sizes and high inventory levels, lean manufacturing promotes smooth flow, pull-based production, and just-in-time (JIT) delivery, where products are made only when needed and in the required quantity. Now a days, lean manufacturing is widely applied not only in large industries but also in small, medium, and craft industries, as well as in services, healthcare, and logistics. Its low-cost, people-centric approach makes it a powerful tool for improving productivity, quality, and competitiveness in modern manufacturing systems.

1.2 Origin

Early ideas of manufacturing efficiency were documented by Frederick Taylor and applied by Henry Ford at the Ford manufacturing plant in the early 1900s, focusing mainly on physical efficiency rather than management philosophy and statistical control. After World War II Lean's formalized manufacturing efficiency methods and philosophies were introduced to Toyota's Shigeo Shingo and Taiichi Ohno by the father of "Lean Manufacturing" Dr. William Edwards Deming. In 1988 manufacturing efficiencies and philosophies were dubbed 'Lean' by John Krafcik, and then were defined in 'The Machine that Changed the World' and further detailed by James Womack and Daniel Jones in 'Lean Thinking' (1996)[1].

1.2.1 Japan: the origins of Lean

The adoption of just-in-time manufacturing in Japan and many other early forms of Lean can be traced back directly to the US-backed Reconstruction and Occupation of Japan following World War II. Although these methods improved efficiency in U.S. military manufacturing during WWII, they were largely ignored by American industries. After the war, Deming was invited to Japan as a consultant, where his ideas were readily accepted and applied by companies such as Toyota and Mitsubishi[2].

Japan's post-war constraints forced industries to adopt highly efficient production systems. As a result, Japanese manufacturers minimized inventory, built smaller factories, and focused on producing only what was immediately needed. These conditions encouraged the development of Just-In-Time (JIT) and lean practices with Toyota playing a central role. Shigeo Shingo and Taiichi Ohno also redesigned Toyota's manufacturing system. Toyota identified waste and quality problems in its processes and responded by implementing Kaizen (continuous improvement) teams, which eventually evolved into the Toyota Production System (TPS) and The Toyota Way.

1.2.2 Evolution and Present

While early ideas related to lean manufacturing appeared in the 1950s and even earlier in Henry Ford's work, Just-In-Time (JIT) was formally developed and systematized by Toyota as part of the Toyota Production System (TPS). Knowledge of TPS spread to Western countries in the late 1970s and from around 1980, these ideas began to be widely implemented in the United States and other developed nations accelerated by conferences and industry discussions, notably a major 1980 conference at Ford's headquarters.

By the mid-1980s and early 1990s, international case-study books documented JIT implementation across Japan, Europe, the U.S., and Australia, showing its applicability beyond manufacturing to areas such as accounting, design, and systems. During the 1990s, the publication of 'The Machine That Changed the World'[3] helped shift terminology from JIT to lean manufacturing and lean management.

1.3 Objectives

The primary objectives of implementing lean manufacturing in this craft unit are detailed as follows:

i. **Reduction of Material Waste and Optimization of Material Cost:**

Wood is the primary raw material, and the unit uses different types with significant cost differences. Sisau is the most expensive, while Uthis is relatively inexpensive. Currently, wood usage is often based on immediate availability rather than systematic planning. Lean manufacturing aims to standardize the selection of timber according to product specifications, reduce unnecessary cutting and leftover scrap, and optimize storage and handling to prevent damage or loss. These practices align with the lean principle of eliminating non-value-adding activities and ensuring that every resource contributes directly to customer value.

ii. **Reduction of Labor Time Variability**

The difference in fabrication time between skilled and less-skilled workers highlights inconsistencies in production. Lean manufacturing addresses this by implementing standardized work methods, clear work instructions, and continuous improvement practices. The goal is to reduce time variation among workers, improve predictability in scheduling, and enable less-skilled workers to achieve performance levels closer to skilled workers. Standardized processes not only ensure reliability but also create a foundation for continuous improvement.

iii. **Better Utilization of Manpower**

Labor costs are a significant portion of total production expenses. Skilled workers are efficient but expensive, while less-skilled workers are slower. Lean principles aim to optimize workforce allocation based on skills and task requirements, implement workflow and process improvements that allow semi-skilled workers to perform effectively, and reduce idle time and unnecessary motion. By aligning tasks with worker capabilities and streamlining operations, the unit can achieve higher productivity while controlling costs.

iv. **Reduction of Overall Lead Time**

The unit experiences delay due to waiting between fabrication, post-fabrication, and outsourced engraving. Lean implementation involves mapping the entire process flow to identify bottlenecks, minimizing waiting and transportation times, and coordinating outsourced activities with internal schedules. These measures implement lean concepts of flow and pull, reducing the total time from raw material to finished product and improving responsiveness to customer demand.

v. Reduction of Dependency on Outsourcing

Engraving is currently outsourced, introducing both cost and time inefficiencies. Lean manufacturing seeks to analyze whether certain tasks can be integrated in-house or, if outsourcing remains necessary, improve scheduling and batch handling to reduce unnecessary waiting or rework. This reduces external dependencies while maintaining product quality, aligning with the lean principle of addressing problems at the source and continuously improving processes.

vi. Improvement of Cost Efficiency

With a fixed selling price of Rs 650 per item, controlling internal costs is critical for profitability. Lean manufacturing focuses on minimizing waste of materials, time, and labor, reducing unnecessary costs from delays and inefficiencies, and increasing output without additional resource consumption. By optimizing production costs, the unit can improve profit margins while maintaining competitive pricing.

vii. Enhancement of Workplace Organization and Workflow

Currently, tools, materials, and workstations may not be optimally arranged, which contributes to motion and time waste. Lean implementation includes applying 5S principles to organize tools, raw materials, and finished products, creating visual management systems for easy identification of materials and tools, and streamlining physical workflow to reduce unnecessary movement. These practices improve operational efficiency, enhance worker safety, and increase process visibility, creating a more structured and productive work environment.

1.4 Achievements Through Lean Manufacturing Implementation

After applying lean principles, the unit is expected to see measurable improvements:

i. Controlled Material Usage and Reduced Waste:

Standardized wood selection and improved cutting practices reduce scrap, while proper storage prevents damage and loss, lowering overall material costs without compromising quality.

ii. **Reduction in Fabrication Time:**

Standardized work instructions allow semi-skilled workers to complete tasks more efficiently. The gap between skilled and less-skilled workers narrows, and production times become more predictable.

iii. **Improved Manpower Productivity**:

Workers utilize time more effectively, idle periods are minimized, and skilled labor is allocated to complex tasks while semi-skilled workers handle standardized processes.

iv. **Reduced Lead Time:**

Process mapping and workflow improvements reduce waiting between stages. Better coordination with outsourcing vendors ensures timely completion of engraving, shortening the overall production cycle.

v. **Optimized Outsourcing Costs:**

Efficient scheduling reduces batch delays, and gradual integration of tasks in-house can decrease reliance on external vendors while maintaining quality.

vi. **Increased Cost Efficiency and Profitability:**

Waste reduction and labor optimization lower production costs per item, increasing net margins and improving competitiveness in the local market.

vii. **Enhanced Workplace Organization and Safety:**

Implementing 5S creates cleaner, more organized workstations. Tools and materials are easily accessible, motion waste is reduced, and a safer, structured environment improves morale and productivity.

1.5 Principles of Lean Manufacturing[4]

1. **Value:**
   At the heart of lean manufacturing is the principle of value. Value is defined as any action or process that directly contributes to meeting customer needs and requirements. Lean organizations strive to identify and prioritize activities that

add value while eliminating those that do not, thereby maximizing efficiency and customer satisfaction.

2. **Value Stream:**

   The value stream represents the sequence of activities required to deliver a product or service to the customer. It encompasses all steps, from raw material acquisition to delivery, including both value-added and non-value-added activities. Lean organizations analyze and optimize the value stream to minimize waste, reduce lead times, and enhance overall process efficiency.

3. **Flow:**

   Flow refers to the smooth and uninterrupted movement of materials, information, and activities throughout the value stream. By eliminating bottlenecks, reducing batch sizes, and standardizing processes, lean organizations strive to achieve continuous flow, enabling faster response times, reduced inventory, and increased productivity.

4. **Pull:**

   The pull principle emphasizes producing only what is needed, when it is needed, and in the quantity needed, based on customer demand. Rather than relying on forecasts or pushing products through the production process, lean organizations implement pull systems that signal the need for production based on actual customer orders or consumption, minimizing excess inventory and waste.

5. **Perfection:**

   Perfection, also known as striving for excellence, is the ultimate goal of lean manufacturing. Lean organizations continuously seek to improve processes, eliminate defects, and exceed customer expectations. While perfection may be unattainable, the pursuit of perfection drives a culture of continuous improvement and innovation.

CHAPTER 2 METHODOLOGY

Lean Manufacturing requires the organization/production line to follow certain generalized procedures in order to analyze the doings carefully in order to best reduce waste and increase value.

These procedures are classified into different categories, each with different characteristics of the production line, customer psychology and waste in order to be evaluate and eliminate the excess. These are further generalized and best explained in terms of the five lean principles, which is one of the ways integrate lean into any manufacturing or production procedure. As with any proper study or analysis, incorporating lean onto any point of production requires for the assessor to define the value of the process and production from the perspective of the customer. Time and resources that are expended without the direct additional value to customer or the final product is one that could be cut or replaced while starting from raw material all the way to the customers. This might include something similar to a company rather outsourcing some parts rather than opening its own shop if the cost, the time and resources of the company would be negatively, leading to customers to pay more for the same end product that now can be had without the extra cost of inhouse manufacturing. Similarly, mapping the value stream allows for all managers, advisers, or any one in general to see any waste in the system more clearly. Checking steps and reordering activities can allow for the waste of the system to be managed well before the production floor or the assembly line has an issue. Birds eye view of the core line of production is very advantageous. A flowchart or visual might allow for view of redundancies and much required common-sense interpretation the paces through the process.

Creating flow follows likewise to the mapping principle. Any production line requiring upstream and downstream coordination makes it unnecessarily redundant to have high production in one singular part of the process while having almost no output in the other. These sets of bottlenecks can be easily removed with the visualization and mapping principle or through simple techniques that are standardized in lean.

Establishing pull is as important as smoothness internally in manufacturing or production. You can’t be selling paper signboards advocating for the stop of deforestation like how you can’t make a book for the illiterate. This means that knowing your customers wants and needs are important. Any changes in the demand are as important so as to be in line with seasons, cultural change, health and safety, and even customer feedback. In simple, managing so that upstream operations produce only to downstream/customer demand.

Finally, striving for excellence is the value that all the levels of production should be motivated to achieve. The five lean principles are like a recipe that gets better with each time it is used. The continuous iteration of them to the manufacturing process, learning from customers, and following best practices. The goal is to create a culture of continuous improvement. When that happens, it becomes a natural part of how things are done.

The primary wastes described in Lean Manufacturing Process are as follows:

Table 1 Wastes of Lean Manufacturing Process

S. No	Waste Type	Definition	Lean Management Tools
1.	Defects	Products that require rework, repair, or are scrapped entirely	**Poka-Yoke** (Error proofing), **Jidoka** (Autonomation), Standardized Work
2.	Overproduction	Making more than is needed, or making it faster than required.	**Kanban** (Pull system), Takt Time (Pace setting), JIT (Just-in-Time).
3.	Waiting	Idle time for workers or machines due to bottlenecks or delays.	**Heijunka** (Leveling), Total Productive Maintenance (TPM), Continuous Flow
4.	Non-utilized Talent	Underusing the skills, ideas, and creative potential of employees	**Kaizen** (Continuous improvement), Cross training, Gemba Walks.
5.	Transportation	Unnecessary movement of materials or products.	Cellular Manufacturing, Spaghetti Diagrams, VSM (Value Stream Mapping).

6.	Inventory	Excess raw materials, WIP (Work-in-Process), or finished goods	Single-Piece Flow, **Kanban**, JIT, Inventory Supermarkets.
7.	Motion	Unnecessary physical movement by people.	5S System, Ergonomic Design, Motion Economy Principles
8.	Excess-Processing	Adding features or steps that the customer doesn't care about.	A3 Problem Solving, Process Mapping, VOC (Voice of the Customer).

2.1 Time and Motion Study

Time and motion study looks closely at how long each task takes and how people and parts move through the workshop, using measures like cycle time (time to finish one piece at a step), takt time (how fast customers need products), and lead time (total time from order to delivery). It helps spot wasted movement and uneven work—for example, one carver waiting because sanding or polishing is slower—so tasks can be balanced across workers or stations.

2.2 Root Cause Analysis

Root cause analysis is used when a defect or delay keeps coming back, so the team fixes the real reason instead of only the symptom. A simple method is 5 Whys (asking “why?” again and again), and a fishbone (Ishikawa) diagram groups causes under Man, Machine, Method, Material, Measurement, and Environment.

2.3 Process Capability

Process capability analysis checks whether a process stays consistent and how much it varies, especially when making many similar items. In a woodcraft shop, this could mean tracking thickness after planning, hole size after drilling, or fit between mortise and-tenon parts; tools like control charts are often used to see if variation is stable and acceptable.

2.4 Kaizen

Kaizen (continuous improvement) means making small, regular improvements with the people who do the work every day, not only managers. In handicraft woodworking, daily kaizen could be moving tools closer to the bench or labeling sandpaper grits, while a short kaizen event could redesign the finishing area to reduce waiting for drying and rework after dust contamination.

2.5 Just-in-time (JIT)

Just-in-time (JIT) aims to make only what is needed, when needed, in the right quantity, so money is not stuck in extra inventory.

2.6 5S process

5S organizes the workplace so tools are easy to find and problems are easy to see (Sort, Set in Order, Shine, Standardize, Sustain)[5].

2.7 Kanban Analysis

Kanban analysis studies how cards/bins/signals control WIP and replenishment and where flow breaks down. For example, if the "ready for finishing" rack overfills while the finishing table is idle, the Kanban limits or replenishment timing can be adjusted to keep work moving smoothly.

2.8 Poka-yoke (error proofing)

Poka-yoke (error proofing) prevents mistakes before they become defects. In a handicraft shop, simple poka-yoke examples include a jig that only allows the correct drilling position, a template that prevents carving beyond a boundary, or a go/no-go gauge that catches incorrect tenon size before assembly.

Figure 1 Lean Manufacturing Wastes

CHAPTER 3 ANALYSIS CONDUCTED IN CASE STUDY

The case study which primarily focused on lean manufacturing processes involved in craft industry, was conducted in technical training centre, Kathmandu University. The TTC manufactured a craft item (token) initially using polymers and wood which was later replaced by entirely wood to minimize the material cost and waste. The total amount of items that were made was 350 units. Several wastes involved in lean manufacturing process was introduced in this process. An example of study of Kathmandu University's Token through lean analysis and methodologies is as follows:

- **Value Stream Mapping (VSM)**

A conceptual value stream analysis of the initial process reveals excessive nonvalue-added time. While actual transformation activities (printing, painting, cutting) were relatively short, the total lead time was dominated by waiting, transportation, and coordination delays. Outsourcing 3D printing introduced long transportation loops and schedule dependency, while internal printing capacity was constrained by the availability of a single printer. VSM highlights that the product spent more time in queues and transit than in value creation.

- **Waste (Muda) Analysis**[6]

Multiple forms of waste were identifiable in the initial process:

  - **Waiting:** Printer downtime, failed prints, and outsourced printing turnaround times. Transportation: Repeated movement between different university facilities and external vendors.
  - **Over-processing:** Use of 3D printing for components that did not require its geometric freedom.
  - **Defects and Rework:** Print failures and surface imperfections requiring repainting or reprinting.
  - **Material Waste:** Discarded support structures and excess filament. This analysis suggested that waste was embedded in both the process design and the product design.

- **Bottleneck and Capacity Analysis**

The single in-house 3D printer functioned as a critical bottleneck. Any failure or delay at this point halted downstream activities. While outsourcing temporarily alleviated capacity constraints, it introduced higher costs and additional waste in the form of transport and coordination. The process lacked robustness and flexibility, a key violation of lean flow principles.

- **Motion and Layout Analysis**

Although woodworking at the technical training centre incurred no direct financial cost, repeated movement of parts and personnel between locations constituted unnecessary motion. Lean analysis recognizes that such motion represents real process waste regardless of accounting treatment.

- **Material Utilization Analysis**

Early designs exhibited low material efficiency. Additive manufacturing generated unavoidable support waste, while wooden components required cutting from larger sections, producing excess scrap. The ratio of useful material to raw material input was therefore low.

3.1 Lean Methodologies and Techniques in Process Improvement

1. **Kaizen (Continuous Improvement)**

   The transition from a predominantly 3D-printed design to a hybrid design and finally to a fully wooden product reflects a Kaizen-driven improvement trajectory. Each iteration addressed specific inefficiencies identified through prior analysis, demonstrating incremental yet cumulative improvement rather than a single radical redesign.

2. **Design for Manufacturability and Assembly (DFMA)**

   Lean DFMA principles were implicitly applied in later iterations. The final design favoured straight cuts, planar surfaces, and laser engraving over complex geometries. This aligned the product with existing workshop capabilities and minimized processing steps, reducing both lead time and error potential.

3. **Just-In-Time (JIT) Thinking**

   While no formal JIT system was implemented, JIT principles were evident in the final approach. Production relied on readily available materials within the workshop rather than procuring new stock. Components were produced only when required, avoiding excess inventory and storage.

4. **Kanban-Like Signalling**

   Informal Kanban behaviour emerged through clear production signals: once materials were selected and engraving tasks assigned, downstream operations proceeded without interruption. The simplified workflow reduced the need for explicit scheduling, as task completion itself acted as a signal for the next step.

5. **Poka-Yoke (Error Prevention)**

   The shift to laser engraving and standardized wooden components reduced opportunities for human error. Unlike painting or multi-part assembly, laser engraving inherently constrained errors in alignment, depth, and text accuracy. The design itself functioned as a Poka-Yoke mechanism by limiting variability.

6. **Standardized Work**

   The final production method consisted of a small number of clearly defined steps that could be executed by one or two skilled workers. This standardization reduced dependency on individual expertise and enabled consistent quality and predictable completion times.

7. **Waste-Minimizing Material Strategy**

   The use of leftover wooden beams or offcuts from other workshop activities significantly improved material efficiency. The design was intentionally structured so that each major component could be produced from a single beam, minimizing scrap generation and improving yield.

3.2 Case Study: The "Token of Love" Project

The team analysed the production of the "Token of Love" for the KU Running Shield, comparing our initial student batch of 50 units with the commercial run of 350+ units at TTC.

Figure 2 Initial prototype of token

Figure 3 Final manufactured token

3.2.1 Design for Manufacturing (DFM): The "Modular" Innovation

A core Lean achievement was our approach to material utilization, which we developed during the first 50 units and TTC continued for the 350 units.

- **The Waste**: Carving complex shapes like the "KU" letters from a single solid block of wood would create massive "Sawdust Waste."

- **Our Solution**: We implemented a Modular Assembly approach. Instead of carving, we cut straight cuboid strips of wood and joined them to form the text structures.
- **Impact**: This maximized material utilization and reduced raw material costs. By using simple, straight cuts to build complex structures, we reduced scrap and saved time.

3.2.2 Phase 1: The Initial State (Our Student Group)

During the production of the first 50 units, we encountered several inefficiencies in the 3D-printing workflow:

- **Process**: We 3D-printed the shields and name tags. These then required Silver Painting to achieve the desired look.
- **The Bottleneck**: This created a massive "Waiting" waste. Each part had to be printed, painted, and then left to dry before assembly. If the paint wasn't perfectly dry, it caused defects during handling.
- **Cost & Quality**: The unit cost was approximately Rs. 450 per piece for logo, and rushing the curing process led to warped 3D parts.

3.2.3 Phase 2: Scale-Up at TTC (Process Optimization)

To scale to 350 units, TTC optimized the workflow by merging steps:

- **Value Engineering**: TTC eliminated the silver painting station. Instead of printing and painting, they used Laser Engraving directly onto the wooden surface.
- **Process Merging**: A significant Lean improvement was that the wood could be finished with *Chapra* first. The laser then engraves the text *through* the finish.
- **Impact**: This eliminated the "Waiting" time associated with silver paint drying. The engraving is the final step, and the token is ready for delivery immediately after.
- **Hybrid Cell Layout**: TTC used a manual expert with a circle-cutting jig for the blanks (High speed) and Laser Engraving for the text (High precision), acting as a Poka-Yoke (error-proofing) mechanism.

Table 2 Comparison table between initial and commercial batch

Metric	Phase 1: Our Initial Batch (50 Units)	Phase 2: TTC Commercial Batch (350 Units)
Construction	Modular (Joined Cuboids)	Modular (Joined Cuboids)
Surface Finish	3D Printing + Silver Paint (Slow drying)	Laser Engraving on Chapra (Immediate)
Cost (Logo)	~Rs. 450 / unit	Rs. 372.83 / unit
Wastes Eliminated	Material Waste (via Modular Design)	Waiting & Defect Waste (via Laser)

3.3 Calculations

The calculations based on the data obtained from Technical Training Centre, Kathmandu University in manufacturing of craft item (token of love) for commercial use are given below:

Obtained Parameters,

Materials Used for making the craft item:

- Salla, (NRs 1400/cubic feet= NRs. 0.05/cm^3).
- Sisau, (NRs 3955/cubic feet= NRs. 0.139/cm^3).
- Utis, (NRs 590/cubic feet= NRs. 0.020/cm^3).

The craft piece consists of three parts that are the circular section at top, the KU structure at middle and the base. The volume of each craft piece is as follows:

1. Volume of Circular Section = 29.25 cm^3.
2. Volume of KU Structure = 64.8 cm^3.
3. Volume of Base = 10.80 cm^3.
4. Total volume of craft piece = 104.85 cm^3

3.3.1 Fabrication Cost:

Table 3 Worker's Cost Table

S. No	Worker's Category	Charges/hr.	Time
1.	Skilled	NRs. 224.28	1 hr.
2.	Normal	NRs. 171.42	2 hrs.

The fabrication cost is primarily based on worker's charges, post-fabrication cost, and cost for laser engraving. The data is tabulated below:

Thus, taking the average charges and manufacturing time for each product:

- Average Charges/hr. = NRs. 197.8
- Average Time = 1.5 hrs.

The post- fabrication cost is NRs. 122.85/hr. and time taken for each product is 1.15 hrs. on average and the cost for laser engraving is NRs. 50/piece.

The widely used wood for making the craft piece (token of love) is Utis. Thus, this calculation is based on wood of type Utis.

The cost of wood per craft piece = $104.85 \times 0.020 = NRs. 2.18$

Thus, the cost of manufacturing each unit of craft piece (Total Cost/unit) = $2.18 + 197.8 + 122.85 + 50 = NRs. 372.83$

Total Number of Units produced (N) = 350.

Work Time (T_w) = 5 months 3 days = 153 days.

Since, a worker works 7 hours per day, the total work time would be:

$T_w = 1071\ hrs.$

Cycle Time (T_c) = 3 hrs.

Hence,

1. $System\ Takt\ Time\ = \frac{T_w}{Number\ of\ good\ parts\ produced} \ldots (1)$

$$= \frac{1071}{350} = 3.05\ hrs.$$

2. $Overall\ Equipment\ Effectiveness\ (OEE) = \frac{Good\ Units \times Cycle\ Time}{Total\ Time} \dots$ (2)

$$= \frac{350 \times 3}{1071} = 0.98$$

3. $Lead\ Time\ = Inventory\ \times System\ Takt\ Time\ \dots$ (3)

$$= 350 \times 3.05 = 1067.5\ hrs.$$

For, Calculating the number of Kanban, we need following parameters:

$$Hourly\ Demands\ (D) = \frac{350}{1070} = 0.32\ units.$$

$$Safety\ Factor\ (S)\ =\ 10\%\ =\ 0.1$$

Number of units per Kanban Card (C)= 15 units. (A container can accommodate 15 units)

4. $Number\ of\ Kanban[6] = \frac{D \times Lead\ Time \times (1+S)}{C} \dots$ (3)

$$= \frac{0.32 \times 1067.5 \times (1+0.1)}{15} = 25.$$

Hence, the number of containers required for entire inventory is 25.

5. $Net\ Production\ Output\ (Before\ Lean\ Manufacturing) =$ $(Retail\ Price - Capital) \times Quantity\ of\ Goods$

$$= (600 - 372.83) \times 35 = NRs. 79500.$$

CHAPTER 4 OBSERVATION AND RESULTS

The case study allowed us to analyze the parameters such as overall equipment effectiveness which was had the best result and resulted into better utilization of equipment in manufacturing process. The system takt time was calculated to be 3.05 hours and the total manufacturing cost per unit was calculated to be Rs. 372.82. This cost however does not include the cost of additional waste of mobility. The number of Kaban was calculated to be 25 with total capacity of each container being 15. It was noticed that most of the part of token of love was made from wooden wastes. Since, the initial 3D- printed parts in the craft item was replaced with entirely wooden assembly, the process followed the concept of Kaizen. The JIT strategy was used in the process of manufacturing which used pull system to manufacture the product whenever the customer demand was available.

However, several recommendations have been made for TTC for future improvement in the manufacturing process. The recommendations are as follows:

4.1 Short-Term Improvements (Current Level)

- **Visual Management (Andon Lights):** Install simple red/green lights on machines. This allows supervisors to see from across the room if a machine is idle or malfunctioning, reducing "Waiting" time.
- **Kaizen (Continuous Improvement):** TTC should have a 5-minute meeting every morning where students/workers suggest one small thing to make the work easier that day.
- **Cross-Training:** Ensure every worker is trained in all stations (Cutting, Laser, and Finishing). This prevents production from stopping if one specific expert is absent.

4.2 Scaling to High Volume (10,000 to 100,000 Units)

If TTC needs to scale to massive quantities, the following Lean steps are necessary to maintain quality and lower costs:

- **Automation of the Bottleneck:** While the hand-machine is fast for 350 units, it will become a "human fatigue" bottleneck at 10,000. TTC should invest in a

CNC Wood Router that can cut 50–100 circular blanks at once from a single large sheet.

- **Multi-Machine Handling:** In Lean, one worker should manage a "cell" of 3 or 4 laser engravers simultaneously. While one machine is engraving, the worker is unloading the second and loading the third. This maximizes Human-Machine Efficiency.
- **Conveyorized Finishing:** For 100,000 units, hand-applying *Chapra* is too slow. TTC should implement a Spray-Finishing Tunnel with a conveyor belt. This ensures the coat thickness is identical every time and uses a heat-lamp drying section to move products to packaging in minutes.
- **Total Productive Maintenance (TPM):** At high volumes, a machine breakdown is a disaster. TTC must implement a schedule where machines are cleaned and calibrated *before* they break, ensuring 100% uptime.
- **Supplier Integration (Just-In-Time):** For 100,000 units, we cannot store all that wood at TTC. They should partner with a timber supplier to deliver pre-sized wood blocks in "Daily Batches" (Kanban) so the workshop space stays clear and organized.

CHAPTER 5 CONCLUSION

Hence, the case study was conducted on lean manufacturing practices that were implemented in small and craft industry. The TTC is a small-scale industry that made craft item such as tokens. Some of the principles of lean manufacturing such as pull system and perfection were well practiced. The production process was not well utilized since wastes such as mobility wastes and wastes incurred by waiting were prominent. The Kaizen concept was well implemented since there had been iterative and gradual improvements in unit's design with time to minimize the wastes to maximum.

Other waste minimization process that could be implemented in TTC is the Kanban process for effective inventory handling and proper signalling system to reduce the waiting time. The JIT principle adopted in TTC reduced the overproduction of tokens and the lead time was approximately equal to cycle time. The mobility waste due to the laser etching process can be minimized by ensuring the availability of laser etching equipment within the TTC.

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